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CASE STUDY:2

TOPIC: URBAN AIR QUALITY IN INDIAN CITIES – EXPLORATORY DATA ANALYSIS AND HYPOTHESIS TESTING

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1. ABSTRACT

Air pollution is a major environmental and public-health challenge in many Indian cities. This case study examines daily air-quality observations from the Central Pollution Control Board (CPCB) for the period 2015–2020. It focuses on the distributions of PM2.5 and nitrogen dioxide (NO_2), differences across cities and seasons, uncertainty estimates, and the change in pollutant levels during the 2020 COVID-19 lockdown compared with the corresponding period in 2019.

The analysis workflow uses Python with Pandas for data preparation, SciPy for statistical tests, Statsmodels for confidence intervals and model-based analysis, and Seaborn for visualization. Descriptive statistics, seasonal summaries, confidence intervals, independent-sample tests (Welch's t-test and/or Mann–Whitney U test), and pollutant correlation analysis are proposed. The report emphasizes that correlation does not establish causation and that lockdown-related changes may also reflect weather, emissions, mobility, monitoring coverage, and other factors.

This is a reproducible analysis plan and report template. Numerical findings are intentionally not fabricated; exact city-wise statistics, confidence intervals, p-values, and conclusions must be generated after running the analysis on the selected CPCB data extract.

2. INTRODUCTION

Urban air quality is influenced by transport, industry, construction, power generation, household fuel use, waste burning, and meteorological conditions. Particulate matter and gaseous pollutants can vary substantially between cities and across seasons. Long-term, city-level monitoring data provide an opportunity to describe these patterns and evaluate whether observed differences are statistically distinguishable.

This case study uses CPCB air-quality observations for Indian cities from 2015 to 2020. It applies exploratory data analysis (EDA) and hypothesis testing to summarize PM2.5 and NO_2, compare seasonal patterns, examine the 2020 lockdown period against a comparable period in 2019, and explore relationships among pollutants.

3. PROBLEM STATEMENT

Air-quality measurements are not necessarily complete or evenly distributed across cities, dates, or monitoring stations. Raw observations may contain missing values, duplicate records, inconsistent station coverage, and extreme readings. A simple comparison of annual or daily averages can therefore be misleading.

The problem is to prepare and analyze CPCB city-day air-quality data in a transparent manner, summarize PM2.5 and NO_2 distributions by city and season, quantify uncertainty using confidence intervals, test whether the 2020 lockdown period differs from the same calendar period in 2019, and interpret pollutant correlations without making unsupported causal claims.

4. OBJECTIVES

- Collect and inspect CPCB city-day air-quality data for 2015–2020.
- Clean date, city, pollutant, and measurement fields and document missingness.
- Summarize PM2.5 and NO_2 distributions by city and season.
- Visualize temporal, seasonal, and city-wise air-quality patterns.
- Calculate confidence intervals for selected means or medians, with assumptions stated.
- Compare pollutant concentrations during a defined 2020 lockdown window with the matching 2019 calendar window.

- Use Welch's independent-samples t-test and/or Mann–Whitney U test as appropriate.
- Measure correlations among pollutants and explain why correlation is not causation.
- Communicate limitations and findings in a concise five-minute briefing.

5. DATASET DESCRIPTION

The proposed source is the Central Pollution Control Board (CPCB), Government of India, using publicly available air-quality monitoring observations. The study period is 2015–2020. The exact download or API extract, its retrieval date, fields, monitoring-station coverage, and licence or reuse conditions should be recorded in the final notebook.

Item	Description
Source	Central Pollution Control Board (CPCB), Government of India
Geographic coverage	Selected Indian cities represented in the downloaded data
Study period	2015–2020
Temporal unit	Daily city-level summary, derived from observations if needed
Main pollutants	PM2.5 and NO_2
Additional variables	Other pollutants and station/date metadata, if available
Analysis type	EDA, confidence intervals, hypothesis testing, correlation

Important data-design note: CPCB releases may be station-level or observation-level rather than pre-aggregated city-day data. If so, city-day values must be created by a documented aggregation rule, such as the daily mean across valid stations. Station coverage and the number of valid measurements contributing to each daily value should be retained where possible.

6. TOOLS AND TECHNOLOGIES

Tool	Purpose
Python	Programming and reproducible analysis
Pandas	Loading, cleaning, grouping, and summarizing data
SciPy	Welch t-test and Mann–Whitney U test
Statsmodels	Confidence intervals and statistical utilities
Seaborn	Distribution, seasonal, time-series, and correlation plots
Jupyter Notebook	Documented, interactive analysis workflow

7. METHODOLOGY

7.1 Data Collection and Scope

Download the relevant CPCB data covering 1 January 2015 through 31 December 2020. Record the source URL or portal name, access date, file name, filters, and any API parameters. Select cities with adequate observations across the comparison windows. Do not treat missing observations as zero.

7.2 Data Cleaning and Preparation

- Parse timestamps into a consistent date/time format and derive date, year, month, and season.
- Standardize city and station names while retaining original identifiers.
- Convert pollutant measurements to numeric values and flag non-numeric or invalid entries.
- Identify missing values, duplicate observations, and implausible readings; document all exclusion rules.
- Check units and ensure that PM2.5 and NO_2 are compared in consistent units.
- Aggregate station-level readings to city-day values only after defining the minimum station/observation coverage rule.
- Keep a record of observation counts so that changes in monitoring coverage can be considered.

7.3 Exploratory Data Analysis

For each city and pollutant, calculate count, mean, median, standard deviation, minimum, maximum, and selected percentiles (for example, 25th and 75th percentiles). Use histograms and box plots to inspect skewness and extreme values. Use line plots to examine daily or monthly trends and seasonal plots to compare distributions.

7.4 Seasonal Classification

For a consistent India-focused analysis, define seasons before examining results. One practical classification is Winter (December–February), Pre-monsoon/Summer (March–May), Monsoon (June–September), and Post-monsoon (October–November). December should be assigned consistently to the winter season, with the season-year handled carefully when grouping across calendar years.

7.5 Confidence Intervals

Calculate 95% confidence intervals for selected city-wise or seasonal means. A standard mean confidence interval can be calculated using the t distribution when its assumptions are reasonably met. Because air-pollution measurements may be skewed and serially correlated, interpret simple daily-observation intervals cautiously. Where appropriate, use bootstrap intervals or aggregate to independent units such as city-day or city-month before estimating uncertainty.

A confidence interval expresses uncertainty around an estimated parameter under the chosen sampling and modelling assumptions; it is not a range containing 95% of individual daily readings.

7.6 Lockdown Comparison: 2020 vs 2019

Define the lockdown comparison window explicitly before testing. A commonly used national lockdown window is 25 March–31 May 2020; compare it with 25 March–31 May 2019 for the same cities and, where possible, comparable monitoring stations. The selected window and rationale must be stated in the final submission. Also consider a shorter strict-lockdown period as a sensitivity analysis if the project scope permits.

For each city and pollutant, formulate H0: the 2020 and 2019 comparison-period distributions or means are equal; H1: they differ. Use Welch's two-sample t-test for a difference in means when observations are independent and variance equality is not assumed. Use the Mann–Whitney U test when a rank-based

comparison is more appropriate. The Mann–Whitney test is not simply a test of medians unless distributional shapes are comparable.

Use a significance level such as alpha = 0.05. Report the test statistic, p-value, sample sizes, effect size or estimated difference, and confidence interval. If many cities or pollutants are tested, address multiple comparisons (for example, Benjamini–Hochberg false-discovery-rate control). Daily observations may be autocorrelated; where possible, use city-day matched comparisons, block bootstrap, or a model that accounts for time dependence.

7.7 Pollutant Correlation

Calculate Pearson correlation for approximately linear relationships and Spearman rank correlation for monotonic relationships or data affected by skewness and outliers. Use paired observations only and report the number of paired records. Visualize the correlation matrix using a heatmap and inspect scatter plots for selected pollutant pairs.

Correlation does not imply causation. Two pollutants may move together because they share emission sources, weather conditions, seasonal cycles, or traffic patterns. A third variable may influence both. Correlation alone cannot establish that one pollutant causes a change in another.

8. PYTHON IMPLEMENTATION

The following example demonstrates a reusable workflow. Update the input file name and column names to match the actual CPCB download.

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from scipy import stats
from statsmodels.stats.multitest import multipletests

# Load the CPCB extract (adjust file and column names as required)
df = pd.read_csv("cpcb_air_quality_2015_2020.csv")

# Standardize column names
df.columns = df.columns.str.strip().str.lower()

# Example expected fields: city, date, pm2.5, no2
df["date"] = pd.to_datetime(df["date"], errors="coerce")
for col in ["pm2.5", "no2"]:
    df[col] = pd.to_numeric(df[col], errors="coerce")

df["city"] = df["city"].astype("string").str.strip()
df = df.dropna(subset=["city", "date"])
df = df.drop_duplicates()

# Create calendar fields
df["year"] = df["date"].dt.year
df["month"] = df["date"].dt.month

def season(month):
    if month in [12, 1, 2]:
        return "Winter"
    if month in [3, 4, 5]:
        return "Pre-monsoon"
    if month in [6, 7, 8, 9]:
        return "Monsoon"
    return "Post-monsoon"

df["season"] = df["month"].apply(season)
```

```
# City-season descriptive statistics
summary = (
    df.groupby(["city", "season"])[["pm2.5", "no2"]]
      .agg(["count", "mean", "median", "std"])
)
print(summary)

# 95% t-based confidence interval for a sample mean
def mean_ci(series, confidence=0.95):
    x = series.dropna()
    n = len(x)
    if n < 2:
        return pd.Series({"n": n, "mean": x.mean(),
                          "ci_low": np.nan, "ci_high": np.nan})
    mean = x.mean()
    sem = stats.sem(x)
    low, high = stats.t.interval(
        confidence, df=n-1, loc=mean, scale=sem
    )
    return pd.Series({"n": n, "mean": mean,
                      "ci_low": low, "ci_high": high})

ci_summary = (
    df.groupby("city")["pm2.5"]
      .apply(mean_ci)
)
print(ci_summary)

# Example lockdown comparison: 25 March to 31 May
# Compare matching calendar dates in 2019 and 2020.
start_2019, end_2019 = "2019-03-25", "2019-05-31"
start_2020, end_2020 = "2020-03-25", "2020-05-31"

a = df.loc[df["date"].between(start_2019, end_2019), "pm2.5"].dropna()
b = df.loc[df["date"].between(start_2020, end_2020), "pm2.5"].dropna()

# Welch's independent-samples t-test
if len(a) >= 2 and len(b) >= 2:
    t_stat, p_value = stats.ttest_ind(a, b, equal_var=False)
    print("Welch t-test:", t_stat, p_value)

# Mann–Whitney U test
if len(a) > 0 and len(b) > 0:
    u_stat, u_p = stats.mannwhitneyu(
        a, b, alternative="two-sided"
    )
    print("Mann-Whitney U:", u_stat, u_p)

# Correlation between PM2.5 and NO2
paired = df[["pm2.5", "no2"]].dropna()
print("Pearson:", stats.pearsonr(paired["pm2.5"], paired["no2"]))
print("Spearman:", stats.spearmanr(paired["pm2.5"], paired["no2"]))

# Visualizations
sns.boxplot(data=df, x="season", y="pm2.5")
plt.title("PM2.5 Distribution by Season")
plt.xticks(rotation=20)
plt.tight_layout()
plt.show()

sns.heatmap(paired.corr(method="pearson"), annot=True, cmap="coolwarm")
plt.title("Pollutant Correlation")
plt.tight_layout()
plt.show()
```

9. HYPOTHESIS TESTING SUMMARY

Analysis	Null hypothesis (H0)	Method	Report
2020 vs 2019 PM2.5	No difference in the defined comparison period	Welch t-test and/or Mann–Whitney U	Statistic, p-value, effect size, CI, n
2020 vs 2019 NO_2	No difference in the defined comparison period	Welch t-test and/or Mann–Whitney U	Statistic, p-value, effect size, CI, n
Pollutant association	No monotonic/linear association	Pearson and/or Spearman correlation	Correlation coefficient, p-value, paired n

Interpret p-values together with effect sizes, confidence intervals, sample sizes, and data quality. A statistically significant result does not automatically imply a large or practically important effect.

10. RESULTS AND DISCUSSION

The completed notebook should present city-wise and seasonal PM2.5 and NO_2 tables, distribution plots, time-series plots, confidence intervals, lockdown comparison tests, and pollutant correlation results. For each test, state the selected dates, city coverage, sample sizes, missing-data treatment, test statistic, p-value, effect size, and conclusion at the chosen significance level.

Do not claim that lockdown caused any observed reduction solely from a before-and-after comparison. Changes may also be associated with meteorology, seasonal conditions, traffic and industrial activity, monitoring-station availability, and broader changes in human activity. The result should be described as an observed difference during the selected period unless a stronger causal design is used.

Numerical results are not inserted in this report because the actual CPCB data file has not been provided or analyzed here. Populate the result tables and conclusions after running the notebook.

11. LIMITATIONS

- Monitoring-station coverage may differ by city and year.
- Missing observations may not be random and can bias summaries.
- Station-to-city aggregation can hide local variation.
- Daily pollution values can be temporally autocorrelated.
- Weather and seasonal changes can confound year-to-year comparisons.
- The lockdown period is not a randomized experiment.
- Multiple hypothesis tests increase the chance of false-positive findings.
- Correlation analysis cannot establish causal relationships.

12. SDG CONTRIBUTION

SDG 11 – Sustainable Cities and Communities

The analysis can help describe urban environmental conditions and support evidence-based discussion of air-quality management, sustainable transport, and healthier urban planning.

SDG 13 – Climate Action

The study supports environmental monitoring and climate-action awareness by examining pollution patterns and the relationship between human activity and urban air quality. Air pollution and greenhouse-gas emissions are not identical, so the report does not treat pollutant concentrations as direct measures of climate change.

13. FIVE-MINUTE BRIEFING / PRESENTATION PLAN

Time	Content
0:00–0:30	Introduce the urban air-quality problem and SDG relevance.
0:30–1:15	Describe CPCB data, cities, years, pollutants, and cleaning.
1:15–2:15	Show city-wise and seasonal PM2.5/NO_2 distributions.
2:15–3:15	Explain the 2019 vs 2020 comparison window, hypotheses, and tests.
3:15–4:15	Present pollutant correlation and explain correlation ≠ causation.
4:15–5:00	Summarize limitations, findings, and implications for urban policy.

14. CONCLUSION

This case study presents a structured approach to exploring urban air quality in Indian cities using CPCB data from 2015–2020. It combines data preparation, descriptive statistics, seasonal analysis, confidence intervals, hypothesis testing, and pollutant correlation analysis. A carefully defined comparison of the 2020 lockdown period with the corresponding 2019 period can identify statistical differences, while appropriate interpretation avoids unsupported causal claims.

The study demonstrates how reproducible statistical analysis can contribute to environmental understanding and evidence-informed urban planning. Its usefulness depends on transparent data selection, consistent aggregation, suitable statistical assumptions, and clear reporting of uncertainty and limitations.

15. REFERENCES

1. Central Pollution Control Board (CPCB), Government of India. National Air Quality Monitoring / air-quality data portal. Use the exact dataset page or download URL accessed for this study and record the access date.
2. Open Government Data Platform India. https://www.data.gov.in/ (Search for CPCB air-quality datasets and verify the selected dataset's coverage and metadata.)
3. SciPy Documentation. Statistical functions. https://docs.scipy.org/doc/scipy/reference/stats.html
4. Statsmodels Documentation. https://www.statsmodels.org/
5. Pandas Documentation. https://pandas.pydata.org/docs/
6. Seaborn Documentation. https://seaborn.pydata.org/
7. World Health Organization. Air pollution. https://www.who.int/health-topics/air-pollution
8. United Nations. Sustainable Development Goals 11 and 13. https://sdgs.un.org/goals